

ASHLEY LIU

+1 (774)4238926 | xil277@ucsd.edu | ashleyliu923@gmail.com

EDUCATION

University of California, San Diego - M.S., Computer Engineering (EC79)

Expected 12/2026

Coursework: *Data Structures & Algorithms, Machine Learning, Deep Generative Models, NLP, Web Mining & Recommender Systems, AI Agents, Computer Architecture, Computer Vision*

Lancaster University & Beijing Jiaotong University - Dual B.S./B.E. in Computer Science, Major GPA 3.76

09/2020 - 06/2024

WORK EXPERIENCE

Software Engineer | AI/ML Intern, California ISO (CAISO), Folsom, CA

06/2026 - 09/2026

- Built and shipped CTRACC, an end-to-end NERC PER-005 training and qualification platform using JavaScript, PHP, and SQL Server, translating trainer and operator needs into task, evaluation, course, and role-based workflows.
- Architected versioned qualification workflows, immutable checkoff records, and RBAC around 30+ audit-critical business rules, enabling historical reconstruction and cross-module traceability.
- Built a deterministic retrieval engine using BM25F, four-lane Reciprocal Rank Fusion, and confidence gating across 120+ approved, versioned knowledge articles for reliable user-facing search.
- Hardened production reliability through load testing, session/IIS tuning, database validation, and deployment checks; authored a 26-file migration specification for a planned React rebuild.

Software Development Engineer, Advanced Micro Devices, Boxborough, MA

04/2025 - 10/2025

- Built an internal Inventory Control System for APU/GPU lifecycle management with Vue 3, Spring Boot, REST APIs, MyBatis, and MySQL, owning RBAC, audit logging, data validation, ticketing, and asset traceability; supported deployment with NGINX and CI/CD.
- Extended KitChecker and QuickEPR across C#/VB.NET codebases while building C++/Python automation pipelines for Linux and Windows, improving test coverage, execution efficiency, and end-to-end workflow reliability.
- Debugged next-generation GPU issues across drivers, system stability, and performance using Linux, DirectX, Vulkan, OpenGL, and AMD profiling tools.

Technical Consultant, AX Future Academy, Boston, MA

12/2024 - 04/2025

- Developed and maintained the academy website using Java/Spring-based frameworks, building backend services, data-driven features, and cloud-hosted deployment workflows; supported AI-assisted search and content discovery features to improve usability and engagement.

Image Processing Engineer Intern, Zuoyouming Technology Co., Ltd, Hubei, CN

07/2023 - 09/2023

- Built an end-to-end PyTorch U-Net segmentation pipeline over 4,340 UAV orthophotos for water-boundary and river-adjacent road extraction; collaborated with urban-planning and environmental stakeholders to translate domain requirements into preprocessing, annotation, augmentation, training, evaluation, and post-processing workflows.
- Engineered a reusable YOLOv5 detection and inference pipeline for riverside buildings, including Labelme/COCO conversion, dataset validation, batch inference, and model tuning with learning-rate tuning, regularization, and transfer learning to improve robustness across varying terrain, lighting, and object scales.

Android Remote Control Application Developer Intern, Hikeen Technology Co., Ltd, Shenzhen, CN

07/2022 - 09/2022

- Built a user-facing Android TV remote-control application with a modular UI, networking, and command-processing architecture, implementing UDP broadcast discovery, IP parsing, and session establishment for phone-to-TV connectivity.
- Designed a lightweight real-time control protocol with sequencing, acknowledgments, retry logic, and asynchronous socket handling, delivering reliable low-latency gesture and command interactions under unstable network conditions.

PROJECT EXPERIENCE

Agentic Chart Reasoning System | Multi-Agent AI / Multimodal Reasoning

03/2026 - 05/2026

- Built a Python multi-agent multimodal reasoning system on the OpenAI API, coordinating task routing, specialized solvers, critics, and verification agents.
- Designed evidence-grounded verification and targeted repair mechanisms to detect inconsistent results and recover from agent failures.
- Improved GPT-4o-mini exact-match accuracy on DVQA hard tasks from 36.7% to 70.0% through modular decomposition and verification.

PUBLICATIONS AND SELECTED AWARDS

- **Liu, X.;** Tao, H. “Java-Based Development of a Vehicle Speed Recommendation Model for Assisted Driving.” IEEE Conference on Machine Learning, Algorithms and Artificial Intelligence (MLAAI), 2023. (Paper ID: MLAAI-2368)
- **1st Prize**-National Univ. Intelligent Vehicle Competition (Vision); Outstanding Graduation Project (Multimodal Lip Recognition, 2023 – 2024); Academic Scholarship (2022 – 2023); Outstanding Student (2020 – 2021); Global Leadership Award (10/2022)

SKILLS

Languages: Python, JavaScript, Java, SQL, C++, C#

Full-Stack & Data: React, Vue 3, Flask, Node.js, Spring Boot, REST APIs, MySQL, SQL Server, JPA, MyBatis

AI/ML: PyTorch, TensorFlow, OpenCV, LLMs, RAG, Information Retrieval, NLP

Infrastructure: Linux, Docker, NGINX, Git, CI/CD